# Anna-Drishti — Research Dossier

Background, current status, evidence base, novelty and team mapping.

Smart India Hackathon 2026 · Internal Review · VIT

---

**HOW TO USE THIS IN THE REVIEW**

Your review is scored on **novelty, research backing, the problem's current status, background, and team inclusion.** This document covers all five, in that order, and each part is written so you can speak from it directly.

The most important section is **Part 3 — the causal chain.** Research backing is not scored by how many citations you list. It is scored by whether the evidence is *load-bearing*: whether removing a source would break your argument. Part 3 shows that every link in our reasoning has a source underneath it.

**CONFIDENCE MARKING — READ BEFORE QUOTING ANY NUMBER**

Two marks appear throughout:

VETTED — from the twelve references in our submitted proposal. The team has already checked these. Quote them freely.

CONFIRM — established storage-science or regulatory facts added here for completeness, but **not** in the original reference list. They are almost certainly correct, but regulatory limits in particular change. **Do not state these as precise figures in the review unless someone has checked them first.** Say "on the order of" instead.

## PART 1 Background — how this problem arises

The narrative to open with. Physical, then biological, then institutional.

### 1.1 · THE PHYSICAL SETTING IS BAGS, NOT SILOS

Indian public grain storage is dominated by **bagged stacks in godowns**, not bulk silos. Grain sits in jute or HDPE bags stacked on dunnage, often several metres deep. This single fact drives everything else, because a bag stack differs from a silo in three ways that matter:

- **Different diffusion behaviour.** Interstitial air paths run between bags as well as between kernels.
- **Different access.** You cannot auger-sample the centre of a bag stack. Probing damages bags.
- **Different instrumentation economics.** Silo monitoring assumes a fixed, permanently instrumented vessel. A godown has many stacks that are built, moved and dismantled.

Nearly all commercial grain-monitoring technology was designed for the silo case. That is the gap this project sits in.

### 1.2 · HOW A SPOILAGE POCKET FORMS

Grain typically enters storage at or near safe moisture. Localised elevation then arises from ordinary, common events:

- Monsoon roof leaks or wall seepage
- A consignment loaded slightly wet into one corner of a stack
- Condensation driven by day–night temperature gradients
- Moisture migration through the stack driven by natural convection

Once a pocket crosses threshold, a **positive feedback** begins. Higher moisture raises respiration; respiration releases both heat and water; the added heat and water raise respiration further. This is why spoilage in grain is not gradual and linear — it accelerates, and by the time it is externally visible it has usually been running for weeks.

### 1.3 · WHY THE INTERIOR IS EFFECTIVELY INVISIBLE

Three failures compound. All three are documented, not assumed.

| FAILURE | WHAT THE EVIDENCE SAYS | SOURCE |
|---|---|---|
| **Sampling misses what matters** | FAO guidance is explicit that insects are not uniformly distributed — they concentrate in dust pockets and zones of heating — so peripheral spear sampling can miss a pocket entirely. The same guidance notes that absence of live adults in a sample does *not* indicate absence of infestation, because eggs and larvae develop concealed inside kernels. | FAO VETTED |
| **Detection thresholds arrive too late** | In humid tropical conditions an infestation as light as one insect per 100 kg can develop into a damaging population within roughly four months. The window for cheap intervention closes long before anything is visible. | FAO VETTED |
| **Heat travels slower than the damage** | Grain is a good thermal insulator, so a developing hot-spot conducts its signal to a temperature probe slowly and weakly. Metabolic $CO_2$ moves with air currents through interstitial space far faster than heat conducts. | Storage-technology literature VETTED |

## 1.4 · WHY THIS MATTERS MORE THAN TONNAGE LOST

Quantity loss is the obvious cost. It is not the important one.

Storage conditions determine **mycotoxin risk**. Storage fungi such as *Aspergillus flavus* require a minimum water activity in the region of **0.82–0.85** to grow, and aflatoxin production occurs within a *narrower* band of conditions than growth itself, with optimum production temperatures reported around **25–30 °C**. VETTED These are precisely the conditions that arise inside a warm Indian godown when moisture migrates into a localised pocket.

> THE SENTENCE THAT FRAMES THE WHOLE PROJECT
>
> **Aflatoxin contamination is not reversible by drying or aeration.** Once the toxin is produced, the grain's fate is decided. That makes this problem inherently **predictive**: the only effective intervention happens *before* the fungus grows. A system that reports damage after the fact is solving the wrong problem.

Aflatoxin B1 is classified by IARC as a **Group 1 human carcinogen** CONFIRM, and both Indian (FSSAI) and export-market regulations set maximum limits for cereals, with EU limits materially stricter than domestic ones CONFIRM. This gives the problem a food-safety and a trade dimension on top of the economic one — but **check the current numeric limits before quoting any.**

## PART 2 Current status — why this is still unsolved

The judging criterion here is whether the problem is live *now*. Four independent lines of evidence say it is.

### 2.1 · THE LOSS IS DOCUMENTED, RECENT, AND OFFICIAL

India's post-harvest cereal losses were assessed at **3.89 % to 5.92 %** in the NABCONS study commissioned by the Ministry of Food Processing Industries and **reported to Parliament in 2024**. VETTED

Why this citation is strong: it is recent, it is government-commissioned, and it was placed before Parliament. Against national foodgrain production measured in hundreds of millions of tonnes, even the lower bound represents millions of tonnes lost annually, a significant share of it during storage.

### 2.2 · THE EXPOSURE IS EXPANDING, NOT SHRINKING

Under the **World's Largest Grain Storage Plan in the Cooperative Sector**, godown capacity is being built at the level of Primary Agricultural Credit Societies, with **over 500 PACS identified for construction** and a national cooperative computerisation programme of **₹2,925 crore** already underway. VETTED

> THE POINT TO MAKE OUT LOUD
>
> Storage capacity is being distributed to thousands of small, locally managed facilities that will have **neither instrumentation nor trained storage scientists on site**. The monitoring gap is not static — it is about to multiply. We are not proposing a solution to a shrinking problem.

### 2.3 · THE CURRENT INTERVENTION IS ACTIVELY DEGRADING

Fumigation in Indian warehouses is largely **calendar-driven rather than condition-driven**: stacks are treated on a schedule regardless of whether that stack needed it, while a stack that did need attention may wait. VETTED

The consequence is not merely waste. Repeated blanket phosphine exposure selects for resistance, and resistance in Indian field populations is now documented and highly variable by geography. A published assessment of Indian *Tribolium castaneum* populations reported resistance ratios ranging from about **2-fold to nearly 71-fold** relative to a susceptible reference strain, with lethal-concentration values spanning more than an order of magnitude between locations. VETTED

Uniform dosing against a non-uniform resistance landscape produces sub-lethal exposures — which is precisely the condition that accelerates resistance further. There is also **no routine verification that a fumigation succeeded**: a leaking gas-proof sheet and a resistant population produce the same outcome, and it is generally discovered weeks later.

### 2.4 · THE GAP IS SPECIFIC, AND NOBODY HAS FILLED IT

| EXISTING APPROACH | WHY IT FALLS SHORT |
|---|---|
| Visual inspection and spear sampling | Peripheral and sparse; misses concealed life stages and interior pockets; damages bags |
| Temperature cables and probes | Heat conducts slowly through grain, so the signal arrives late and attenuated; typically deployed in silos, not bag stacks |

| EXISTING APPROACH | WHY IT FALLS SHORT |
|---|---|
| Calendar-based prophylactic fumigation | Treats stacks that do not need it, misses stacks that do; drives resistance selection; no efficacy verification |
| Laboratory mycotoxin assay | Accurate but destructive, slow, expensive and post-hoc — it confirms loss rather than preventing it |
| Imported commercial monitoring (e.g. TeleSense, OPI) | Technically capable but designed and priced for large Western silo and elevator operations; oriented to bulk storage rather than bagged stacks, and economically out of reach for a PACS-level facility |

STATE THE GAP IN EXACTLY THESE TERMS

There is no affordable, non-destructive, interior-resolved, predictive monitoring system designed for the **bag-stack storage geometry** that dominates Indian public and cooperative warehousing.

## PART 3 The evidence base — our causal chain

This is the core of the research-backing score. Every link in our argument, and the evidence holding it up. If a judge probes any single step, this is where you answer from.

1 **Indian public storage is predominantly bagged stacks in godowns**

**Evidence:** FCI and state warehousing practice; the PACS expansion is building more of the same. **Establishes:** the geometry we must design for.

↓

2 **A bag stack's interior cannot be adequately inspected from outside**

**Evidence:** FAO guidance on inspection of bagged grain — non-uniform insect distribution, peripheral spear sampling, concealed life stages. **Establishes:** the detection problem is structural, not a matter of trying harder.

↓

3 **Localised moisture pockets form and self-amplify**

**Evidence:** established grain-storage physics — respiration is a strong function of moisture and temperature, and releases both. **Establishes:** why a small initial fault becomes a large one.

↓

4 **Above a water-activity threshold, mycotoxigenic fungi grow**

**Evidence:** *A. flavus* requires $a_w$ of roughly **0.82–0.85** minimum for growth. **Establishes:** there is a *specific, measurable threshold* — which is what makes prediction possible at all.

↓

5 **Toxin production occupies a narrower window than growth**

**Evidence:** published temperature/water-activity work on deleterious fungi and mycotoxin production during grain storage; optimum production around **25–30 °C**. **Establishes:** we must model two thresholds, not one — growth, then toxin.

↓

6 **Aflatoxin contamination is irreversible**

**Evidence:** drying or aeration does not remove toxin already produced. **Establishes:** the intervention *must* be predictive. This is the logical hinge of the entire project.

↓

7 **$CO_2$ is an earlier indicator than temperature**

**Evidence:** $CO_2$ is recognised in the storage-technology literature as an earlier and more responsive indicator of spoilage onset than temperature; supported by published $CO_2$-sensor development work for grain quality monitoring. **Establishes:** our sensing choice. Gas moves through interstitial air; heat crawls through insulating grain.

↓

8 **Interstitial humidity at equilibrium approximates grain water activity**

**Evidence:** established moisture-sorption isotherm relationships for cereals — modified Chung–Pfost / modified Henderson forms with commodity-specific coefficients (ASAE D245). **Establishes:** we can obtain $a_w$ from a cheap humidity sensor rather than a laboratory instrument.

↓

9 **Insect activity has a detectable acoustic signature**

**Evidence:** published spectral signatures — near **355 Hz** for *Rhyzopertha dominica*, near **440 Hz** for *Sitophilus oryzae*, and lower-frequency content near **176** and **434 Hz** for *Tribolium castaneum*; demonstrated feasible with inexpensive hardware. **Establishes:** the pest arm of the system is grounded, not speculative.

↓

10 **Therefore: interior $CO_2$ + water-activity sensing enables intervention before the threshold is crossed**

This conclusion is not an assumption. It is what remains once links 1–9 are accepted — and each of those has a source.

**HOW TO USE THIS IF A JUDGE CHALLENGES ONE STEP**

Do not defend the whole chain at once. Ask which step they doubt, then answer that step with its source and stop. A focused answer to one link demonstrates command of the material far better than restating everything.

## PART 4 Novelty — what is genuinely new

Novelty is judged against prior art. Be precise about which parts are new and which are established — claiming novelty for something standard is the fastest way to lose credibility.

### 4.1 · HONEST SEPARATION

| COMPONENT | STATUS | NOTE |
|---|---|---|
| $CO_2$ as a spoilage indicator | Established | Well documented. We do not claim to have discovered this. |
| Sorption isotherms for $a_w$ | Established | Published equations and coefficients, used as intended. |
| Acoustic insect detection | Established | Published signatures and feasibility. |
| Predictive mycology growth models | Established | Standard cardinal-value approach. |
| **Interior-resolved sensing for bag stacks** | NEW | Commercial monitoring targets bulk silos. Bag stacks have different diffusion behaviour and access constraints. The probe lance is designed for that geometry. |
| **Single-analyser multiplexed $CO_2$** | NEW | One analyser sequenced across ports by pump and rotary manifold. Cuts cost, removes inter-sensor drift so depth-to-depth comparison is made by the same instrument against the same baseline, and makes port count expandable without proportional cost. |
| **Biological prediction, not physical measurement** | NEW | Output is a modelled *time-to-risk* derived from growth kinetics, not a temperature or humidity reading. Converts monitoring into prevention. |
| **Closed-loop verification of intervention** | NEW | Post-treatment $CO_2$ decay and acoustic silence confirm efficacy — addressing a failure mode current practice does not detect at all. |
| **Acoustic coupling through the lance body** | NEW | Chosen specifically to reject the ambient noise of a working godown, rather than assuming a quiet laboratory. |
| **In-house labelled validation dataset** | NEW | Sensor time-series paired with fungal and insect ground truth. Genuinely difficult for another team to replicate without laboratory access. |

> THE ONE-SENTENCE NOVELTY CLAIM
>
> The individual signals are established science. **What is new is the combination applied to bag-stack geometry, the single-analyser architecture that makes depth-resolved gas mapping affordable, and the shift from reporting a measurement to issuing a dated, evidence-backed prediction.**

### 4.2 · TECHNOLOGIES DELIBERATELY EXCLUDED

Worth stating, because it demonstrates a problem-first rather than technology-first approach:

- **Distributed ledger.** An append-only hash-chained log is sufficient, auditable and computationally trivial. Adding a blockchain here would be an unjustified use of the technology.
- **Per-depth dedicated $CO_2$ sensors.** Superseded by the shared-analyser manifold.

- **Large language models.** No role in this decision path.
- **Computer vision on stack exteriors.** By definition observes the symptom we are trying to anticipate.

Judges frequently reward teams who can explain what they chose *not* to build. It signals engineering judgement rather than enthusiasm.

## PART 5 Team inclusion — discipline mapped to problem

The criterion is not “is the team diverse” but ”does each discipline do necessary work”. Ours does.

> THE FRAMING SENTENCE
>
> Grain spoilage is a **biological** process, detected through **physical** sensing, and interpreted by **computational** models. This problem was selected in part because it requires precisely the disciplines this team already holds.

| MEMBER | DISCIPLINE | OWNS WHICH PART OF THE RESEARCH |
|---|---|---|
| **Chiranjib Dash** 24BCE2411 | CSE | System architecture and integration; the decision engine that converts interpreted signals into a ranked, dated action |
| **Mohit Kumar** 24BCE2378 | CSE | Transport physics and numerical modelling; signal processing for the acoustic pest pipeline |
| **Vishal Vivek** 24BCE2377 | CSE | Embedded firmware and sensing pipeline; noise modelling, anomaly detection, tamper-evident logging |
| **Krittika Parida** 23BBT0193 | Biotechnology | **Mycology lead.** Water-activity thresholds, fungal growth kinetics, spoilage-timeline validation. Owns links 4, 5 and 6 of the causal chain |
| **Netra R** 24BBT0055 | Biotechnology | **Entomology and grain quality.** Insect ground-truth trials, moisture and assay protocols, and the inspection-baseline model. Owns link 2 and link 9 |
| **Devavrath Ramesh** 24BME0205 | Mechanical | Probe-lance design and fabrication, pneumatic sampling manifold, heat and moisture transport modelling. Owns links 1, 3 and 7 |

### THE ARGUMENT TO MAKE

A team of only computer scientists could build the electronics and the dashboard, but could not generate the biological ground truth that makes the predictions trustworthy. Two Biotechnology members give this project a **wet-laboratory validation capability that a conventional software team cannot reproduce**, and that is the single strongest defensible advantage in our proposal. VETTED

> SAY THIS IF ASKED WHY THE TEAM IS COMPOSED THIS WAY
>
> “When a judge asks where a number came from, the person who owns it answers — not the person who typed it. Every biological threshold in our model is defended by Krittika, every baseline assumption by Netra, and every transport constant by Devavrath. That division is not cosmetic; it is why we can defend the model rather than just run it.”

## PART 6 Reference list

The twelve sources from the submitted proposal, annotated with what each one is actually doing for us. Cite by number in the deck.


1. Ministry of Food Processing Industries / NABCONS post-harvest loss assessment, as reported to Lok Sabha, 2024 — cereal losses 3.89–5.92 %. `downtoearth.org.in/governance/as-told-to-parliament-august-6-2024-4-8-grains-5-15-fruits-vegetables-lost-after-harvest`
   → Establishes scale and currency of the loss. Part 2.1.
2. Press Information Bureau — *World's Largest Grain Storage Plan in the Cooperative Sector*. `pib.gov.in/PressReleasePage.aspx?PRID=2146718`
   → Establishes that exposure is growing. Part 2.2.
3. FAO — *Towards Integrated Commodity and Pest Management in Grain Storage,* Section 6: Inspection and detection methods for storage insect pests. `fao.org/4/x5048e/x5048E0d.htm`
   → The backbone of our inspection-baseline model. Links 2 and 9. **Netra's primary source.**
4. FAO — *Code of Practice for Storage and Fumigation of Bagged Grain*. `fao.org/4/x5048E/x5048E`
   → Bag-stack storage practice and fumigation norms. Part 2.3.
5. *Influence of Temperature and Water Activity on Deleterious Fungi and Mycotoxin Production during Grain Storage*. PMC. `pmc.ncbi.nlm.nih.gov/articles/PMC5780356/`
   → Water-activity thresholds and the growth-versus-toxin distinction. Links 4 and 5. **Krittika's primary source.**
6. *Current status of phosphine resistance in Indian field populations of Tribolium castaneum*. Scientific Reports. `nature.com/articles/s41598-023-43681-y`
   → The 2-fold to 71-fold resistance range. Part 2.3.
7. *Determining the sound signatures of insect pests in stored rice grain using an inexpensive acoustic system*. Food Security. `link.springer.com/article/10.1007/s12571-024-01493-6`
   → Species spectral signatures and low-cost feasibility. Link 9.
8. Mankin et al. — *Automated Applications of Acoustics for Stored Product Insect Detection, Monitoring, and Management*. Insects. `mdpi.com/2075-4450/12/3/259`
   → Establishes acoustic detection as a mature field, not a novelty claim.
9. *Acoustic Detection of Insects in Stored Products in the Presence of Strong Ambient Noise*. Sensors. `mdpi.com/1424-8220/26/5/1511`
   → Justifies contact coupling through the lance body rather than open-air microphony.
10. *Monitoring carbon dioxide concentration for early detection of spoilage in stored grain* — $CO_2$ as a leading indicator relative to temperature.
    → **The single most load-bearing citation in the project.** Link 7. If one source has to be defended, it is this one.
11. *Development of carbon dioxide ($CO_2$) sensor for grain quality monitoring*. Computers and Electronics in Agriculture. `sciencedirect.com/science/article/abs/pii/S1537511010000917`
    → Establishes $CO_2$ sensing in grain as practically achievable.
12. *Optimizing phosphine applications for insect control in bagged rice stored in Indian warehouses*. Journal of Stored Products Research.
    → Indian bag-stack fumigation practice specifically. Part 2.3.


**IF YOU HAVE TIME BEFORE THE REVIEW**

Open references 3, 5 and 10 and read the abstract of each. Those three carry links 2, 4–6 and 7 respectively — between them they hold up most of the causal chain. Being able to say "that paper found X" rather than "we have a citation for that" is the difference between a good and an excellent research-backing score.

## PART 7 Delivering this in the review

### 7.1 · ORDER TO PRESENT IN

| # | COVER | IN ONE LINE |
|---|---|---|
| 1 | Background | Bag stacks are opaque; damage is invisible until irreversible |
| 2 | Current status | 3.89–5.92 % loss, reported to Parliament in 2024; exposure growing via PACS |
| 3 | Why unsolved | Every existing approach is peripheral, late, post-hoc, or priced for Western silos |
| 4 | Research backing | Walk the causal chain — ten links, each with a source |
| 5 | Novelty | Signals are established; the bag-stack application and the architecture are not |
| 6 | Team | Each discipline owns specific links of that chain |

### 7.2 · QUESTIONS YOU WILL GET, AND WHO ANSWERS

| QUESTION | ANSWER WITH | WHO |
|---|---|---|
| “Isn’t $CO_2$ monitoring already done?” | Yes — in bulk silos, in the West, at a price point irrelevant to a PACS godown. We do not claim the signal is new. We claim the geometry, the architecture and the affordability are. | Chiranjib |
| “How do you know 0.82 is the right threshold?” | Published *A. flavus* cardinal values; it is a range, 0.82–0.85, and we use the conservative end. | Krittika |
| “Why should $CO_2$ beat temperature?” | Gas moves with air currents through interstitial space; heat conducts through grain, which is an insulator. Over the same 40 cm our model gives roughly one day versus fifteen. | Mohit or Devavrath |
| “Is your comparison baseline fair?” | It is FAO-grounded. And the key point is not that inspection is incompetent — it is that a spear reaches 0.4 m and the pocket is 1.6 m in. | Netra |
| “What is actually built versus proposed?” | Working simulation with real physics and a real decision engine; ESP32 + SHT3x hardware reading real water activity. $CO_2$ analyser and acoustics are proposed, not yet sourced. | Chiranjib |

**THE ONE HABIT THAT RAISES A RESEARCH SCORE**

When you cite something, say **what it found**, not that it exists. “A 2023 study of Indian *Tribolium* populations found resistance ratios from 2-fold to 71-fold” scores; “we have references on phosphine resistance” does not.

**CLOSING LINE FOR THE RESEARCH SECTION**

> “Every threshold in our model comes from published literature, and every one of them is owned by a specific member of this team who can defend it. We did not pick numbers that made the demo look good — several of them made it harder, and we kept them because they are what the science says.”